Construction and Verification of Long Hairpin RNA Plasmids for Recombinant siRNA Production in *E.coli*

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Abstract

The small interfering RNA (siRNA) was recently found to be an important factor in RNAi (RNA interfering), as it can specifically bind with target sequences on mRNA, and result in the inactivation of mRNA expression. This gives us a possibility to find new methods in treating cancer. We can find certain siRNA to bind with the mRNA of oncogene to silence its expression, which inhibit the further development of cancer cells. The result of such inhibition can be reflected by the morphological structure of host cell after treated by siRNA. Also, to collect enough amount of siRNA, we first inoculate the host cell's target gene into prokaryotic cells (E.coli in this experiment) via plasmid, then siRNA can be amplified in prokaryotic cells. This progress is called "pro-siRNA". In this report, we focus on learning the initial steps in pro-siRNA, and the results we gathered in these steps.

Introduction

RNA interference (RNAi) is a well-established method that allows researchers to selectively reduce gene expression, making it a powerful tool for investigating gene function in health and disease. Small interfering RNAs (siRNAs) are central to this process, but conventional approaches often require careful sequence design and can be costly for large-scale studies. To address these challenges, our lab has developed an innovative system for producing siRNAs directly in bacteria, known as pro-siRNAs. Each pro-siRNA represents a pool of siRNAs naturally processed from a double-stranded RNA, often yielding higher specificity and efficiency without the need for extensive sequence optimization.

The purpose of our SRS project was to apply this pro-siRNA technology to establish a platform for systematic RNAi screening in cancer research. Our long-term goal is to identify genes that are essential for cancer cell survival or proliferation, particularly those that act as "cancer-dependent" genes. Unlike housekeeping genes, which are required by all cells, cancer-dependent genes provide selective vulnerabilities that may be exploited for therapy. By

constructing pro-siRNA libraries and applying high-content imaging analysis, we aim to uncover such candidate genes and generate insights into the cellular pathways that drive tumor growth

Methods

Primers were designed targeting a >200 bp sequence from the gene of interest, avoiding specific restriction sites and off-target sequences (forward with SacI/NotI sites, reverse with XhoI/SalI sites). Then, two DNA fragments (DNA-1: SacI/XhoI; DNA-2: NotI/SalI) were amplified using high-fidelity PCR.

Plasmids and DNA-1 were digested with SacI/XhoI, then purified and ligated using T4 DNA ligase (overnight at 4℃). After that, recombinant plasmids were transformed into E. coli and plated on antibiotic-containing LB agar (incubate overnight). Screen colonies by plasmid extraction, restriction digestion (SacI/XhoI), and sequencing. After that, digest plasmids (with DNA-1 ligated) and DNA-2 with NotI/SalI, repeat the process and screen target plasmid.

Confirm final plasmid by triple digestion (SacI/XhoI, SalI/NotI, SacI/NotI) and Sanger sequencing (hairpin region expected).

Results

Using the pro-siRNA platform, we successfully engineered and validated a panel of full-length long hairpin RNA constructs targeting genes with diverse cellular functions, including *ZNF354A*, *MYO1B*, *GPSM2*, *KMT2E*, *PHF12*, *MICAL3*, *A2M*, *EFHD2*, *EIF4G1*, *GPX4*, *MAPK1*, *NES*, *PAFAH1B1*, *RANBP2*, *SLC4A1AP*, *STT3A*, *and XRCC5X*. For each candidate, a 200-689bp (varies in different genes) region downstream of the ATG start codon was amplified with high-fidelity polymerase to generate sense and antisense fragments carrying distinct restriction sites. Electrophoretic analysis consistently revealed sharp, single bands of expected sizes, confirming reaction specificity. The sense fragments were ligated into the p323 backbone via SacI/XhoI, producing intermediate constructs validated by colony PCR, restriction digestion, and sequencing, which confirmed correct orientation and sequence integrity. Subsequent insertion of the antisense fragments through SalI/NotI ligation yielded full-length hairpin assemblies, with resistant colonies exhibiting digestion patterns precisely consistent with predicted hairpin sizes.

Quality control analyses, including triple restriction mapping and Sanger sequencing, further verified structural fidelity. Sequencing traces across the promoter and coding region were of high quality, with signal attenuation at the SacI junction—indicative of hairpin loop formation—followed by reads corresponding to the reverse complement of the targeted sequences. Collectively, these results establish a verified library of gene-specific long hairpin RNA plasmids that provide a robust molecular toolkit for recombinant siRNA production in *E. coli* and enable systematic interrogation of cancer-relevant gene dependencies in liver cancer cell models.

Discussion

Through this research, we successfully generated partial components of a pro-siRNA library through sequential cloning of DNA1 and DNA2 fragments into the P323 plasmid, which represents not only essential steps within the larger RNAi project, but also a practical training framework that prepared us for future stages of research.

Throughout the experiments, a critical outcome of our work was the refinement of experimental protocols for the existing cloning system. After repetitive troubleshooting, we discovered that extending restriction digestion times from 1.5 h to 2 h noticeably increased digestion efficiency, and that doubling the amounts of both enzymes and DNA substrates improved band visibility and reproducibility in agarose gel electrophoresis. These modifications significantly enhanced the overall success rate of highly potent recombinant siRNAs production in *E. coli*.

While our results verified the successful construction of hairpin RNA plasmids targeting several candidate genes of focus, the contents of our summer project encompassed only a portion of the large project proposal. This project will be continued after the summer vacation, which will then focus on employing the complete pro-siRNA library in a high-throughput RNAi screen and validating top candidate genes by additional siRNA sequences and CRIPSR.

References

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